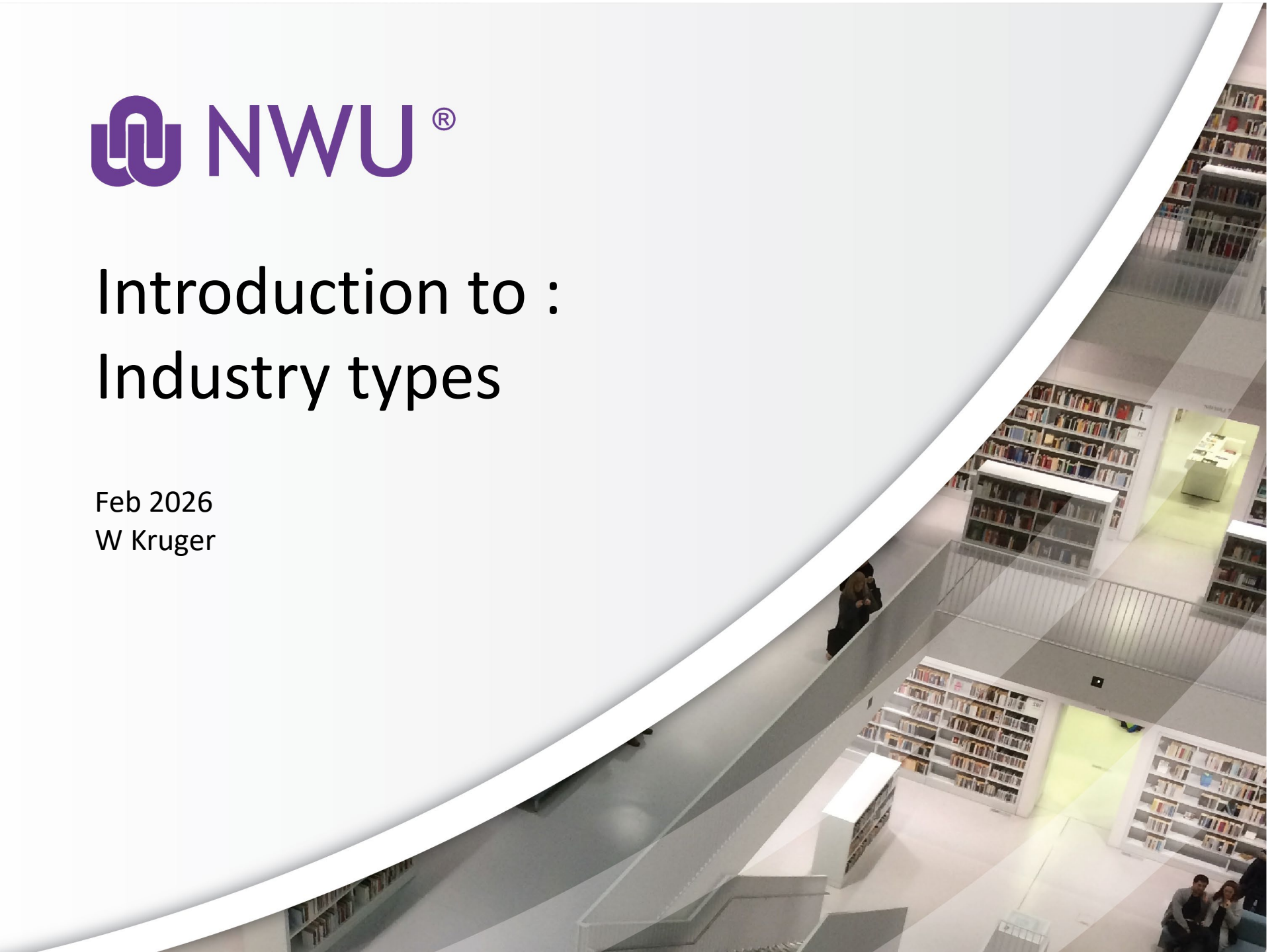


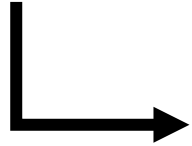


Introduction to : Industry types

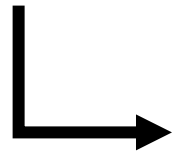
Feb 2026
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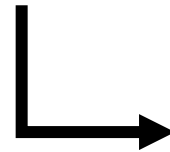
Different
Industries



Different
Purposes



Different
Needs



Different
Designs

Products – transport (airplanes, trains, boats, cars, bikes, bicycles) clothes, food(fresh, cooked, tinned), Equipment (kitchen eg. kettle, oven, washing machine, office eg. computers, printers, stationary) work eg artisan tools, sport eg rackets, balls, hockey sticks), packaging, books, toys etc.

Services – restaurants, cafés, shopping malls, stadiums, events, airports, taxi ranks, administration offices, packaging, cooling, storing facilities, parking lots, schools, universities, homes, flats, sports etc.

Let's have a look at some... *Notice*

- Differences in flows (materials, people and info)
- Interactions of machines vs people
- Design requirements (what are the critical factors for THAT facility)
- How this affects Facility Design

A look at Swartland wooden Doors and Windows factory:

<https://www.swartland.co.za/videos/#gallery-4>

Product manufacturing - Cars



Product manufacturing - Cellphones



Product manufacturing - Food



Services - Hospital



Services - Airport



Services Theme park



Building vs Rebuilding a facility

From scratch = *Green field*

eg. a new home, a
new store, a new plant, a
new site



Upgrading or redefining a
facility = *Brown field*

eg. When a store is sold to
another company, when a
place need to be revamped
or repurposed, improved



A facility affects:

- Operating costs
- Operational effectiveness
- Handling and maintenance costs
- Productivity
- Employee morale
- Flexibility to adapt to customer requirements
- Sustainability over the longterm

So some **BIG** implications!

Some characteristics of designs

- Flexible (if possible)
- Modular
- Upgradable
- Adaptable
- Scalable
- Repeatable
- Environment and energy friendly

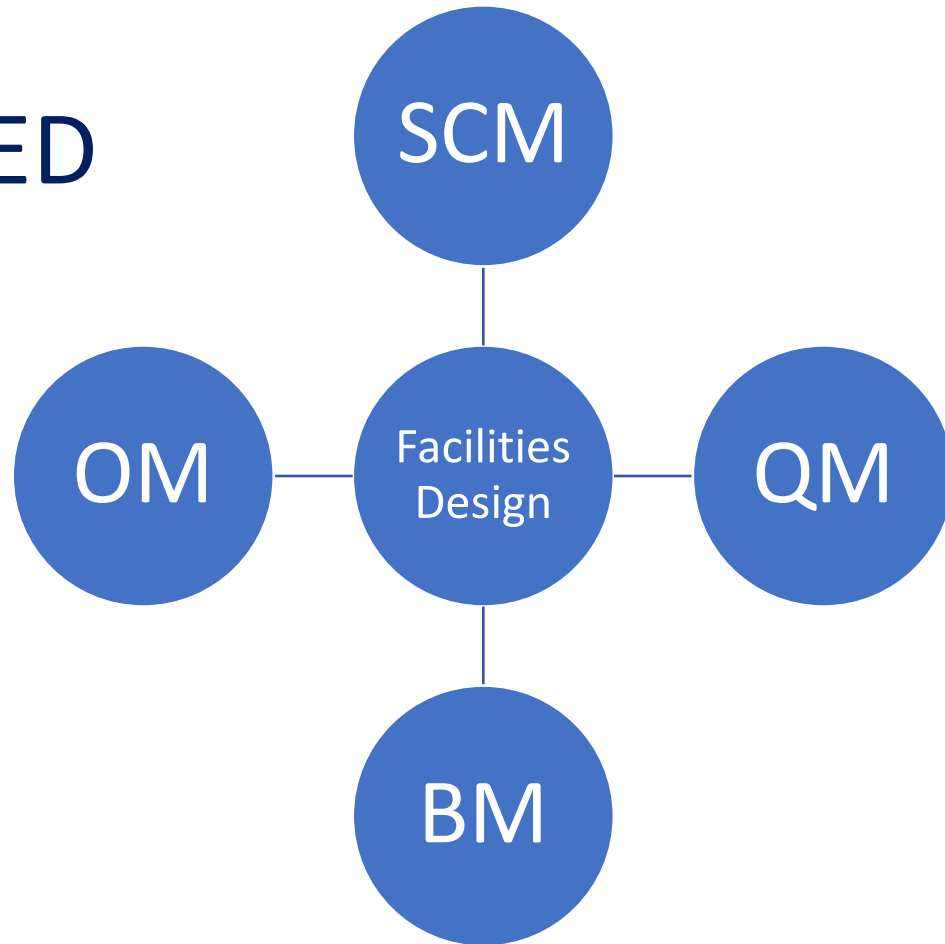
6 key questions to answer before Starting a design...

1. WHAT is produced / or service to be given?
2. HOW ...
3. HOW MUCH... (capacity planning)
4. WHERE
5. WHEN...
6. For HOW LONG...

Refer Ch2 textbook

Facilities design is an

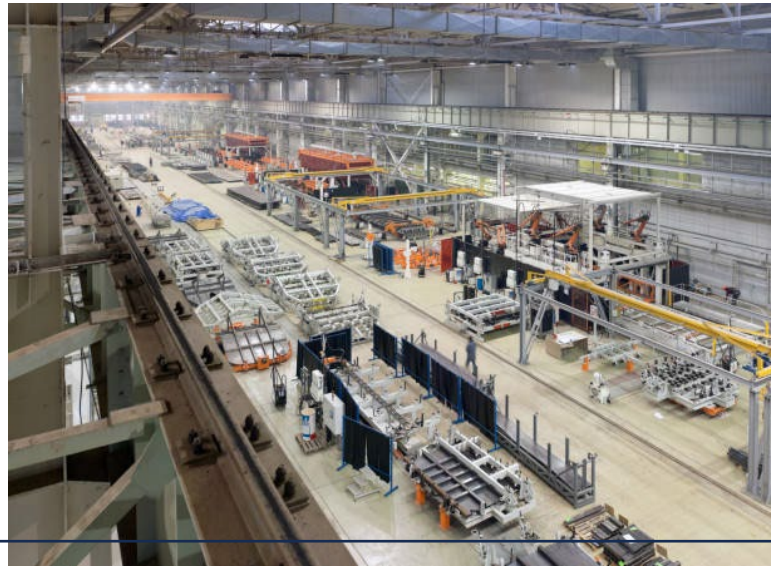
INTEGRATED
course



In manufacturing the specifics include:



**Product
specs**



Process flow

Production Schedule

A	B	C
X		X
	X	
		X

**Scheduling
specs**



Equipment specs



Product design tools

- Quality Function Deployment (QFD)
- Benchmarking
- Prototyping
- Assembly drawings
- Photos

Process design tools

- Bill of Materials (BOM)
- Route sheets
- Assembly chart
- Operation process flow charts
- Precedence diagrams
- Material Requirements Planning (MRP)

Management and Planning tools

1. Affinity diagram
2. Interrelationship graphs
3. Tree diagrams
4. Prioritization matrices
5. Matrix diagrams
6. Process decision program chart
7. Activity network diagram

Refer textbook
Fig 2.26 p73

also Process Flow charts or

Value Stream Maps (VSM) but MOST NB: you need to clearly understand the production or service process

Bottom line of all tools....

To help you have
ENOUGH
ACCURATE
DATA
to do calculations and work with